2020-2021 SPRING SEMESTER FLUID MECHANICS MIDTERM, Date:26/04/2021

Name: **Surname:** **Student ID:**

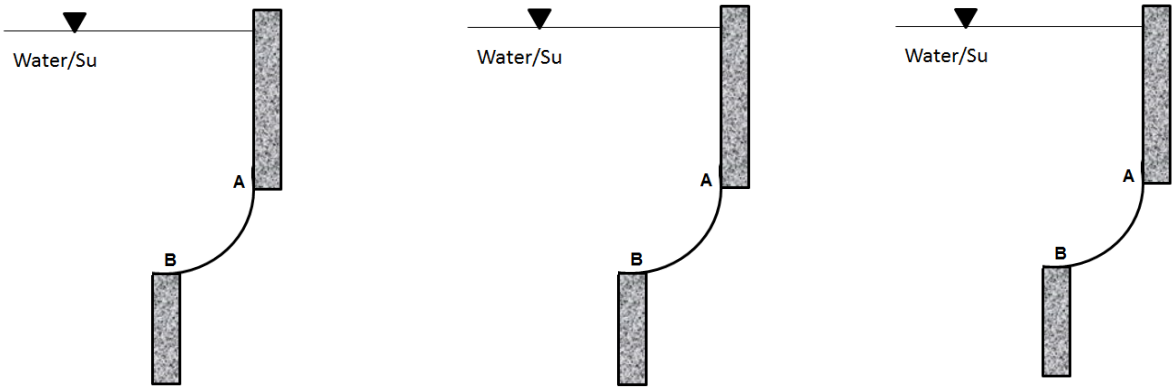
Question 1 (31):

i) (6)

a) Draw the pressure distribution on quarter circle gate AB.

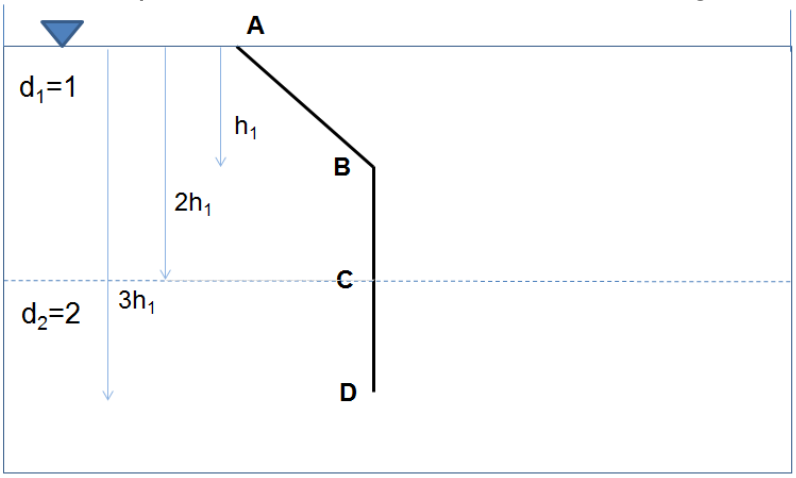
b) Draw horizontal component of the pressure distribution on gate AB.

c) Draw the vertical component of the pressure distribution on gate AB.

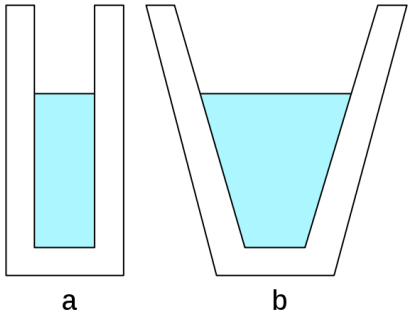


a) b) c)

ii) (6) Draw the hydrostatic pressure distribution on ABCD plane surface shown in the below figure. Write the pressure values A, B, C and D on the figure.



iii) (3) There is water in both containers shown in the figure below. There is 1 liter of water in container "a" and 2 liters of water in container "b". The bottom areas of both containers are 10 cm^2 and the water heights are 20 cm. Determine the hydrostatic forces on the bottom of containers "a" and "b".



iv) (6)

- “$p=\gamma h$” What is the name of this equation?
- Which type of fluid applicable?
- Which type of flow field applicable?

v) (3) If relative pressure is 300 kN/m^2, determine the equivalent water column.

vi) (4) Express the difference between the streamline and the pathline.

vii) (3) Determine the mass of the fluid for 2 m^3 volume, if it has a specific weight of 11200 N/m^3.